Harjeet Singh Chahal

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Education

Rutgers University - New Brunswick, New Jersey, USA Aug 2025 - May 2027

M.S. in Computer Science **GPA: 4.00/4.00**

- **Relevant Coursework**: Machine Learning, Massive Data Mining, Interpretable & Explainable AI, Database Systems for Data Science, Data Structures & Algorithms, Artificial Intelligence, Visual Computing.

Netaji Subhas University of Technology - New Delhi, India Nov 2020 - Jun 2024

B.Tech. in Computer Science & Engineering (Artificial Intelligence) **GPA: 3.74/4.00**

- **Relevant Coursework**: Deep Learning, Natural Language Processing, High Performance Computing, Distributed Computing, Operating Systems, Data Communication, Game Theory, Design and Analysis of Algorithms, Web Technology.

Technical Skills

Languages: Java, Python, C++, Go, SQL, JavaScript, R, HTML/CSS
ML / AI: PyTorch, Scikit-Learn, Hugging Face Transformers, NLTK, LangChain, RAG, OpenCV, NumPy, Pandas
Backend / Frontend: FastAPI, Flask, Node.js, REST, GraphQL, gRPC, MCP, React, React Native, Next.js, Expo
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Linux, Git, CI/CD, SLURM, Prometheus, Grafana
Data Tools: PostgreSQL, pgvector, MySQL, MongoDB, Redis, FAISS, Pinecone, ChromaDB, Kafka

Experience

Rutgers School of Health Professions Piscataway, NJ

Research Assistant : Mobile + ML Engineering Apr 2026 - Present

- Productionized HIPAA-compliant PyTorch EfficientNet-B0 on FastAPI, cutting CPU inference to **25–40 ms**.
- Deployed Dockerized FastAPI ML APIs on **GCP Cloud Run** with **CI/CD** for automated tests, builds, and releases.
- Built an **Expo/React Native** iOS/Android app with JWT auth, reusable UI, deep linking, persistent state, and weather, geolocation, and UV-index APIs; added a context-aware educational chatbot and engagement workflows.
- Built a 50-test Python/FastAPI suite for ML inference, auth, edge cases, and parity; resolved ARM/x86 floating-point drift and regressions and owned **10+ Apple TestFlight releases** tested by **100+ participants**.

Technical Consulting and Research Inc. Weston, CT

Software Development Intern : Backend + AI Engineering Jun 2026 - Aug 2026

- Architected a **local-first** compliance platform (FastAPI, React, PostgreSQL/pgvector) running LLM inference and embeddings on-premise via **Ollama**, keeping client evidence off third-party APIs.
- Built a pipeline restricting citations to retrieved chunk IDs, with server-side source resolution to reduce unsupported claims.
- Developed a **MCP** server exposing 20 read-only compliance tools for agentic retrieval, evidence lookup and control analysis.

Dhruv Research Gurugram, India

Associate : Analytics + Data Engineering Sep 2024 – May 2025

- Engineered Python/SQL ETL pipelines with Pandas/NumPy to clean, validate, and normalize **200K+ survey responses** across 10 state projects, automating QA and transformations to **reduce turnaround time by 30%**.
- Predicted vote share across **500+ constituencies** using Python/SQL, statistical weighting, and MAE/RMSE validation.
- Worked within and coordinated an **8-member team**, using Python/SQL trackers for sampling and demographic skew.

Projects

ScaleTrain : HPC Training Pipeline | PyTorch, DDP, SLURM, NVIDIA A100 | GitHub [May 2026 – June 2026]

- Scaled a 30M-parameter transformer across **4 NVIDIA A100s** using PyTorch DDP + SLURM, sustaining **500K tokens/s**.
- Built fault-tolerant DDP training with **signal-aware checkpoint/restart** and deterministic sharding, resuming from the **exact interrupted step with zero lost work** and bitwise-identical behavior versus uninterrupted runs.

FlexStore : AWS S3-Inspired Object Store | Go, gRPC, PostgreSQL, Docker | GitHub [Sept 2025 – Dec 2025]

- Built a Go object store with 3× replication across 5 nodes, hitting **528 MiB/s writes** and **2.6 GiB/s reads**.
- Engineered heartbeat failure detection and a crash-safe PostgreSQL repair queue, re-replicating 44–50 lost chunks in **8–13 s** after node failure while sustaining **100% read availability** and byte-identical recovery.
- Enforced SHA-256 on every read/write and auto-healed corrupt replicas, validated through fault-injection tests.

SentryFlow : Real-Time API Rate Limiting & Usage Analytics | Go, Kubernetes | GitHub [June 2025 – Aug 2025]

- Engineered and deployed a Kubernetes API rate-limiting platform on AWS, serving **1000+ requests/day at <100ms latency**; implemented **token-bucket/sliding-window controls**, JWT auth, Kafka ingestion, and **92% test coverage**.
- Built a **React** dashboard for real-time API usage and latency analytics, improving **observability and debugging**.

Achievements

Programming: Solved 1,550+ LeetCode problems with a 1,300+ day active streak. | leetcode.com/u/harjeetchahal
Academic Distinction: Ranked top 0.15% nationally among ∼1M candidates in JEE Main 2020; GRE Quant: 166/170.